Notebook 3 — Chapter 3.2

Market Microstructure & Order Book Fair Value

20 worked examples — fair value synthesis, bid/ask/spread/mid, inventory risk & fill probability — calibrated for Stage 2 Finals market-making intuition

A. Fair Value Fundamentals — Fast Mental EV

Example 1 — The Baseline: a Single Die

A contract pays $\$X$ where X is one fair die roll. Find the fair value.

Solution. $\mathbb{E}[X] = \dfrac{1+2+\cdots+6}{6} = \boxed{\$3.50}$

⋆ *Intuition:* every fair-value question is "compute $\mathbb{E}$[payoff] fast." This is the atomic building block — memorize 3.5 for one die outright, you'll use it constantly.

Example 2 — Linearity Beats Enumeration

A contract pays the *sum* of two independent fair dice, in dollars. Find the fair value without listing all 36 outcomes.

Solution. $\mathbb{E}[X_1 + X_2] = \mathbb{E}[X_1] + \mathbb{E}[X_2] = 3.5 + 3.5 = \boxed{\$7.00}$

▶ *Speed trick:* linearity needs no independence and no enumeration — under a 10-second mental clock, never build the 36-cell table when the sum of means is instant.

Example 3 — Scenario Tree

A fair coin picks the regime: Heads → payoff $\sim$ Unif($\$0, \$10$) (mean \$5); Tails → fixed \$2. Find the fair value.

Solution.

$$\mathbb{E}[X] = 0.5(5) + 0.5(2) = 2.5 + 1 = \boxed{\$3.50}$$

⋆ *Intuition:* condition on the branch, take the mean within each branch, then average branches by their probability — the "law of total expectation," the single most-used tool in live fair-value calls.

Example 4 — Weighted Multi-Outcome Contract

A contract pays \$100 w.p. 0.3, \$40 w.p. 0.5, \$0 w.p. 0.2. Find the fair value.

Solution.

$$\mathbb{E}[X] = 100(0.3) + 40(0.5) + 0(0.2) = 30 + 20 + 0 = \boxed{\$50.00}$$

▶ *Speed trick:* compute each term as "round number × probability" left-to-right and running-sum mentally — never write out a table under time pressure if the arithmetic is this clean.

Example 5 — A Non-Round Fair Value

Draw 1 card from a standard 52-card deck. A contract pays \$10 if it's a face card (J, Q, K), \$0 otherwise. Find the fair value.

Solution. 12 face cards out of 52:

$$\mathbb{E}[X] = 10 \cdot \frac{12}{52} = \frac{120}{52} \approx \boxed{\$2.31}$$

⋆ *Intuition:* not every fair value is a clean number — quoting a market still means picking a round bid/ask *around* an ugly fair value (see Section B), not rounding the fair value itself.

Example 6 — Full Linearity Stack

A bundle pays independent contract A (fair value \$4) plus independent contract B (fair value \$6) plus a \$3 bonus if a separate fair coin lands heads. Find the bundle's fair value.

Solution.

$$\mathbb{E}[A + B + 3 \cdot \mathbb{I}_{\text{heads}}] = 4 + 6 + 3(0.5) = 10 + 1.5 = \boxed{\$11.50}$$

▶ *Speed trick:* linearity chains across arbitrarily many pieces, dependent or not — decompose *any* complex payoff into pieces with known means first, add last. This is the one skill every fast-mental-math fair-value round is testing.

B. Bid, Ask, Spread & Mid

Example 7 — Reading a Quote

True fair value is \$50.00. You quote bid \$49.50 / ask \$50.60. Find the spread and the mid, and compare mid to true fair value.

Solution. Spread $= 50.60 - 49.50 = \boxed{\$1.10}$. Mid $= \dfrac{49.50 + 50.60}{2} = \boxed{\$50.05}$.

Mid is \$0.05 *above* true fair value — your quote is skewed, not centered.

⋆ *Intuition:* "mid" is a property of *your quote*, not a measurement of fair value — the two only coincide when you quote symmetrically around your true estimate.

Example 8 — Setting a Symmetric Spread

Fair value \$120.25. You want a total spread of exactly \$0.50, centered on fair value. State your bid and ask.

Solution. Half-spread = 0.25:

$$\text{bid} = 120.25 - 0.25 = \boxed{\$120.00}, \quad \text{ask} = 120.25 + 0.25 = \boxed{\$120.50}$$

▶ *Speed trick:* always compute the *half*-spread first and add/subtract it symmetrically — halving at the start avoids the common error of adding the full spread to one side only.

Example 9 — Spread Should Track Uncertainty

Your fair-value estimate for a contract has std. dev. $\sigma = \$2$ (you're not fully sure of fair value itself). House convention: half-spread $= 0.75\sigma$. Fair value \$75. Quote the market.

Solution. Half-spread $= 0.75(2) = 1.5$:

$$\text{bid} = \boxed{\$73.50}, \quad \text{ask} = \boxed{\$76.50}$$

⋆ *Intuition:* spread width is compensation for *estimation* uncertainty, not just a fixed house edge — a contract you're unsure about deserves a wider market than one you've nailed down exactly, even at the same fair value.

Example 10 — Best Bid / Best Ask Across Makers

MM1 quotes 99.80 / 100.20. MM2 quotes 99.90 / 100.10. A buyer arrives. Who do they trade with, and at what price?

Solution. Best (lowest) ask across makers $= \min(100.20, 100.10) = \boxed{\$100.10}$, offered by **MM2**. The buyer lifts MM2's offer.

▶ *Intuition:* "best bid/best ask" (NBBO) is a market-wide property, not a single maker's quote — the tightest maker wins the flow on each side independently; MM1 could still win the *sell* side if their bid were higher.

Example 11 — Adverse Selection, First Look

True (eventual) fair value is \$50. You quote 49.00/51.00 (wide, \$2.00 spread). A sharp counterparty only buys from you at \$51 when *their own* estimate exceeds \$51.50. On those fills, what's your average realized loss versus your quoted edge?

Solution. You sold at \$51 believing fair value was \$50 (apparent edge +\$1). But fills only happen when the true value is already ≥ 51.50 on average: realized loss $\approx 51.50 - 51.00 = \boxed{\$0.50 \text{ per fill}}$, not a profit.

⋆ *Intuition:* a fill is not a random sample — it's conditioned on someone wanting to trade with you, which is exactly when your price is most likely wrong. This selection effect is the seed of Section C's inventory logic.

C. Inventory Risk & Fill Probability

Example 12 — Skewing After a Fill

Flat inventory, fair value \$100, symmetric quote 99.50/100.50 (± 0.50). You get filled on your bid for 10 units (now long 10). House rule: skew both quotes down \$0.20 per 10 units long. New quotes?

Solution. Shift center from 100.00 to $100.00 - 0.20 = 99.80$, keep the same ± 0.50 half-spread:

$$\text{bid} = \boxed{\$99.30}, \quad \text{ask} = \boxed{\$100.30}$$

▶ *Intuition:* skewing moves *both* quotes the same direction (down, if long) — it makes you more eager to sell (tighter effective ask) and less eager to buy more (worse effective bid), nudging inventory back toward flat.

Example 13 — A Toy Fill-Probability Model

Fill probability on a resting quote: $\mathbb{P}(\text{fill}) = 0.5 - 0.25d$, where d is the quote's distance from fair value in dollars (floored at 0). You place your bid \$1 below fair value. Find $\mathbb{P}(\text{fill})$.
Solution. $d = 1$:

$$\mathbb{P}(\text{fill}) = 0.5 - 0.25(1) = \boxed{0.25}$$

⋆ *Intuition:* every toy fill-probability model encodes the same trade-off — quote closer to fair value (small d) to fill more often, but each fill is worth less edge. There's no free lunch; Example 14 makes this explicit.

Example 14 — Expected P&L of a Two-Sided Quote

Using Example 13's model symmetrically: bid at $d = 1$ (edge \$1 if filled, $\mathbb{P} = 0.25$), ask at $d = 1$ (edge \$1 if filled, $\mathbb{P} = 0.25$), flat inventory. Find expected P&L per round.
Solution.

$$\mathbb{E}[\text{P\&L}] = 0.25(1) + 0.25(1) = \boxed{\$0.50}$$

▶ *Speed trick:* expected P&L from quoting is always $\sum(\text{edge}_i \times \mathbb{P}(\text{fill}_i))$ — compute each side independently and add, exactly like Section A's linearity.

Example 15 — Net of Inventory Holding Cost

Same \$0.50 expected trading P&L per round as Example 14, but you're carrying 5 units of inventory, costing \$0.05/unit/round in volatility risk. Find the *net* expected P&L.
Solution. Holding cost $= 0.05 \times 5 = 0.25$:

$$\text{Net } \mathbb{E}[\text{P\&L}] = 0.50 - 0.25 = \boxed{\$0.25}$$

⋆ *Intuition:* gross quoting edge is not the scoreboard — a market maker who quotes profitably but never flattens inventory is quietly bleeding risk cost every round. This is why Example 12's skew-to-flatten instinct exists at all.

Example 16 — The Reservation Price Formula

Fair value \$100, you hold $q = +8$ units, risk-aversion $\gamma = 0.01$, payoff variance $\sigma^2 = 4$. Simplified reservation price: $r = \text{FV} - q\gamma\sigma^2$. Find r and quote $\pm\$0.50$ around it.
Solution.

$$r = 100 - (8)(0.01)(4) = 100 - 0.32 = \boxed{\$99.68}$$

$$\text{bid} = \boxed{\$99.18}, \quad \text{ask} = \boxed{\$100.18}$$

▶ *Intuition:* this is Example 12's ad-hoc "skew down when long" rule, now with an actual formula — the skew size scales with inventory q, risk-aversion γ, *and* variance σ^2. A volatile contract gets skewed harder per unit of inventory than a calm one.

D. Capstones

Example 17 — Full Quote Construction from a Multi-Outcome Contract

A contract pays \$0 / \$50 / \$100 w.p. 0.2 / 0.5 / 0.3. House convention: half-spread $= 0.02\sigma$. Find the fair value, σ, and the quoted market.
Solution. $\mathbb{E}[X] = 0(0.2) + 50(0.5) + 100(0.3) = 25 + 30 = \boxed{\$55}$. $\mathbb{E}[X^2] = 0 + 2500(0.5) + 10000(0.3) = 1250 + 3000 = 4250$. $\text{Var}(X) = 4250 - 55^2 = 4250 - 3025 = 1225 \Rightarrow \sigma = 35$. Half-spread $= 0.02(35) = 0.70$:

$$\text{bid} = \boxed{\$54.30}, \quad \text{ask} = \boxed{\$55.70}$$

⋆ *Intuition:* this chains Section A (compute $\mathbb{E}[X]$), a variance computation (Ch. 2.2's $\text{Var} = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$), and Section B's quoting rule into one pipeline — exactly the sequence a live market-making round demands.

Example 18 — Sequential Quoting with Evolving Inventory

Round 1: fair value \$100, flat, quote 99.5/100.5, filled on the *ask* for 1 unit (now short 1). Round 2: fair value moves to \$101. Skew rule when short: $r = \text{FV} + |\text{short}| \times \$0.30$. Quote $\pm\$0.50$ around r.
Solution. $r = 101 + 1(0.30) = \boxed{\$101.30}$:

$$\text{bid} = \boxed{\$100.80}, \quad \text{ask} = \boxed{\$101.80}$$

▶ *Intuition:* short ⇒ skew *up* (mirror image of Example 12's long ⇒ skew down) — raising both quotes makes your bid more eager to buy back the short, at the cost of a worse (higher) ask.

Example 19 — Fast Mental Math via an Identity

A spinner shows $1, 2, 3, 4$ with $\mathbb{P}(k) = k/10$. A contract pays $\$k^2$. Find the fair value using the fastest possible mental shortcut.
Solution.

$$\mathbb{E}[X] = \sum_{k=1}^{4} k^2 \cdot \frac{k}{10} = \frac{1}{10}\sum_{k=1}^{4} k^3$$

Use $\sum_{k=1}^{n} k^3 = \left(\sum_{k=1}^{n} k\right)^2$: here $\sum k = 10$, so $\sum k^3 = 100$.

$$\mathbb{E}[X] = \frac{100}{10} = \boxed{\$10.00}$$

⋆ *Intuition:* "sum of cubes = (sum)^2" is exactly the kind of buried identity that turns a 40-second computation into a 5-second one — under a Stage 1/2 time bound, recognizing the pattern is worth more than raw arithmetic speed.

Example 20 — Full Capstone: Variance Amplifies the Skew

Contract pays \$0/\$60/\$120 w.p. 0.25/0.5/0.25. You're long $q = 4$, $\gamma = 0.005$. Find the fair value, variance, reservation price, and a $\pm\$3$ quote around it.
Solution. $\mathbb{E}[X] = 0 + 30 + 30 = \boxed{\$60}$. $\mathbb{E}[X^2] = 0 + 1800 + 3600 = 5400 \Rightarrow$ $\text{Var} = 5400 - 3600 = \boxed{1800}$.

$$r = 60 - (4)(0.005)(1800) = 60 - 36 = \boxed{\$24.00}$$

$$\text{bid} = \boxed{\$21.00}, \quad \text{ask} = \boxed{\$27.00}$$

▶ *Full-circle intuition:* the reservation price dropped \$36 off a \$60 fair value — not because the inventory $q = 4$ is large, but because $\sigma^2 = 1800$ is huge. Example 16's formula makes this explicit: skew scales *linearly* in variance, so a high-variance multi-outcome contract punishes inventory far harder than a calm one at the same position size. This is the entire chapter in one number.

One-tool-tree reminder: fair value is always $\sum$(outcome × probability), computed fast via linearity and identities (Section A); a quote is a spread placed around an estimate, widened for uncertainty (Section B); inventory and fill probability pull that center away from fair value toward a reservation price that protects you from adverse selection (Section C). Everything in Section D is just these three ideas, chained.